Reg. No. | | | | | | | | | | | [17AE46]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2019

ENGINEERING MATERIALS AND MANUFACTURING PROCESSES

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed.

PART - A

1.	**a.**	Write a note on classification of Titanium alloys. Also list the properties, composition and applications of each grade.	**08**
	b.	Briefly explain the grades of steel with respect to their application and compositions.	**06**
2.	**a.**	What are Plastics? Differentiate Thermoplastics with Thermosettings with examples and properties.	**06**
	b.	What are different types of glasses? What are its ideal properties and applications in aircraft?	**08**
3.	**a.**	What are Patterns? Write a note on Pattern materials and Pattern allowances.	**08**
	b.	Explain Centrifugal casting with a neat sketch. List the advantages.	**06**
4.	**a.**	What is Pressure die casting? Explain a Hot chamber pressure die casting with a neat sketch.	**07**
	b.	What are Casting defects? Explain any six Casting defects with reasons and possible remedies.	**07**

PART - B

5.	**a.**	What is the difference between Fusion and Pressure welding? Explain Ultrasonic welding with a neat sketch? List the advantages and applications of ultrasonic welding.	**08**
	b.	Explain Thermit welding with a neat sketch. List its applications.	**06**
6.	**a.**	Explain the following forging processes with neat sketches: Swaging, Upsetting, Heading, Fullering.	**08**
	b.	Explain Thread rolling and Ring rolling operations with neat sketches.	**06**
7.	**a.**	Compare Direct and Indirect extrusion processes with neat sketches.	**08**
	b.	What is a Drawing ratio? Explain a Wire drawing machine with a neat sketch.	**06**

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8. The given drawing, in Fig.8 is a part to be made by cast iron by casting. Determine the pattern size and write the sketch of the pattern after giving allowances. Shrinkage allowance for cast iron is 2%, Machining allowance =2mm on all sides. All dimensions in mm. 50 castings are to be produced. Suggest a suitable pattern material and type of pattern **14**

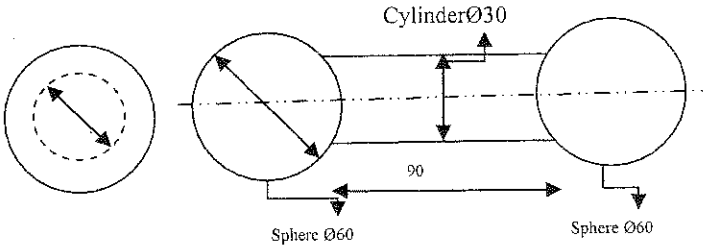

Fig. 8